

Accuracy and speed across resolutions

Poisson 2D, heat 2D, Burgers 2D and reflective waves. Frozen-network baseline configurations.

Median complete GPU query time; worst error over all cases, times and repetitions. N is intervals per axis. FOM/ROM values above one favor the ROM against that named solver.

Problem	N	ROM ms	FOM ms	FOM/ROM	ROM error %
Poisson 2D	64	2.468	5.734	2.324	6.803
Poisson 2D	256	2.355	22.279	9.460	6.802
Poisson 2D	1024	2.495	188.212	75.448	6.802
Heat 2D	64	11.227	5.506	0.490	4.559
Heat 2D	256	11.341	20.997	1.851	4.556
Heat 2D	1024	12.206	210.596	17.253	4.555
Burgers 2D	64	36.226	78.240	2.160	10.85
Burgers 2D	256	35.747	117.473	3.286	4.551
Burgers 2D	1024	41.677	605.747	14.534	3.908
Reflective waves	64	4,510.116	110.527	0.025	3.641
Reflective waves	256	4,530.336	135.462	0.030	3.569
Reflective waves	1024	4,532.594	1,053.343	0.232	3.568

Problem	Primary FOM	Displayed error norm	Cases / repeats
Poisson 2D	CG, tolerance 1e-06	Current-relative solution	30 / 3
Heat 2D	CN-CG, tolerance 1e-6	Current-relative field	12 / 3
Burgers 2D	Newton 1e-06 / linear 1e-08; FFT	Fixed-initial-relative field	4 / 3
Reflective waves	Midpoint CG, tolerance 1e-06	Current-relative displacement	2 / 3

Burgers accepts stall exits under its recorded stopping rule; stationarity is unmeasured. Reflective-wave qualification checks velocity and energy-state errors as well as displacement. Different error norms cannot be ranked across PDEs.

These are development results with final cases unopened. Training, mesh assembly and compilation are excluded; supplied-input processing, solves and full-field device outputs are included. Pair timings within a job, never across different GPUs.

Poisson: online speed–accuracy tuning

Same frozen network. Baseline and FOM controls are retimed together in a new Poisson allocation.

Fastest stationary NMROM at $N = 1024$: $32.004494\times$ faster than the fastest tested passing CG FOM.

ROM 2.613982 ms versus 83.659173 ms for CG 1e-01. ROM worst error: 6.801556%; does not meet the 5% accuracy gate. This is a measured runtime ratio.

17 frozen-checkpoint settings screened at 64 intervals on all 30 Gaussian-source development cases. The fixed shortlist is confirmed at every requested mesh with 3 repeats; endpoint selection remains developmental.

N	Selection	Configuration	ROM ms	Error %	Adjusted %	Stationary	Outliers	5% gate	ROM vs tight CG	ROM vs passing CG
64	Baseline	Baseline; M257	2.361	6.802558	6.802961	90/90	1	fail	2.439 $\times$ faster	1.376 $\times$ faster
64	Fastest tested	Budget 1; M257	1.488	30.069926	30.070417	0/90	1	fail	3.869 $\times$ faster	2.182 $\times$ faster
64	Fastest stationary	Stop 1e-06; M257	2.210	6.802558	6.802961	90/90	0	fail	2.605 $\times$ faster	1.470 $\times$ faster
64	Most accurate*	4 starts; M1024	6.427	6.802474	6.802877	90/90	0	fail	1.116 $\times$ slower	1.979 $\times$ slower
256	Baseline	Baseline; M257	2.616	6.801564	6.801967	90/90	1	fail	9.797 $\times$ faster	4.353 $\times$ faster
256	Fastest tested	Budget 1; M257	1.673	29.999485	29.999975	0/90	2	fail	15.321 $\times$ faster	6.808 $\times$ faster
256	Fastest stationary	Stop 1e-06; M257	2.347	6.801564	6.801967	90/90	0	fail	10.920 $\times$ faster	4.852 $\times$ faster
256	Most accurate*	4 starts; M1025	6.636	6.801553	6.801956	90/90	0	fail	3.862 $\times$ faster	1.716 $\times$ faster
1024	Baseline	Baseline; M257	2.943	6.801556	6.801959	90/90	1	fail	63.140 $\times$ faster	28.428 $\times$ faster
1024	Fastest tested	Budget 1; M257	1.970	29.995107	29.995598	0/90	0	fail	94.335 $\times$ faster	42.473 $\times$ faster
1024	Fastest stationary	Stop 1e-06; M257	2.614	6.801556	6.801959	90/90	0	fail	71.083 $\times$ faster	32.004 $\times$ faster
1024	Most accurate*	4 starts; M1025	6.942	6.801550	6.801953	90/90	0	fail	26.768 $\times$ faster	12.052 $\times$ faster

Tight CG uses tolerance 1e-06; passing CG is the fastest tested iterative FOM meeting the target, with its setting listed below. “Times faster” divides FOM by ROM median time; “times slower” reverses that ratio. All timings are paired within this job. “Fastest tested” includes nonstationary early exits. Adjusted error includes the empirical reference allowance. GPU timing outliers are retained.

No tested configuration meets the original physical and numerical target on every development case. *Most accurate is restricted to settings with every selected solve stationary at the original threshold; an unrestricted accuracy winner is shown separately when different. The largest reduction in worst error from baseline is 0.00008462 percentage points across the confirmed meshes.

N	Tight CG ms	Fast passing CG ms	Fast CG error %	Fast CG setting	Direct DST ms
64	5.759	3.248	1.135	CG 3e-02	0.137
256	25.627	11.387	2.272	CG 1e-01	0.139
1024	185.810	83.659	1.019	CG 1e-01	0.319

Frozen k16/r64 decoder; no retraining or online empirical quadrature. M is the actual retained weak-mode count. These are preassembled, precompiled presets: operators and code caches are prepared for each mesh and M; budgets and gradient thresholds select compiled kernels. New unprepared settings incur offline work. The original stationarity gate is 1e-06; in-loop checks are charged, post-query auditing is excluded for every method.

How the FOM tolerance changes the comparison

Fastest tested iterative FOM passing the same development target and original numerical checks.

Problem	<i>N</i>	Passing FOM	FOM ms	GPU FOM/ROM	Host FOM/ROM
Poisson 2D	64	CG, tolerance 3e-02	3.223	1.306	1.175
Poisson 2D	256	CG, tolerance 1e-01	9.685	4.112	3.223
Poisson 2D	1024	CG, tolerance 1e-01	84.584	33.907	14.901
Heat 2D	64	CN-CG, tolerance 1e-2	2.504	0.223	0.257
Heat 2D	256	CN-CG, tolerance 1e-2	6.939	0.612	0.655
Heat 2D	1024	CN-CG, tolerance 1e-2	59.178	4.848	2.267
Burgers 2D	64	none	—	—	—
Burgers 2D	256	Newton 3e-03 / linear 5e-01; dense DST	13.963	0.391	0.429
Burgers 2D	1024	Newton 1e-02 / linear 5e-01; FFT	68.044	1.633	1.442
Reflective waves	64	Midpoint CG, tolerance 1e-02	82.648	0.018	—
Reflective waves	256	Midpoint CG, tolerance 1e-06	135.462	0.030	—
Reflective waves	1024	Midpoint CG, tolerance 1e-06	1,053.343	0.232	—

Ratios use the baseline ROM on page 1. A ratio alone is diagnostic when that ROM fails its target. Host timing includes input/output transfers where measured; the wave study measured only output transfer, so its complete host ratio is unavailable. All selections use opened development cases.

Direct-transform controls at the largest mesh

Problem	<i>N</i>	Control	GPU ms	Error %
Poisson 2D	1024	Direct sine-transform FOM	0.317	0.0007351
Heat 2D	1024	Direct sine-transform FOM	1.236	0.0003501
Reflective waves	1024	Direct sine-transform FOM	16.624	0

A win against iterative CG does not imply a win against a direct sine-transform solver. These controls retain their original measurement contract. Full control rows, retained timing outliers and raw JSON links are in the adjacent Markdown report.

Accuracy checks and provenance

The tables preserve the existing definitions and stopping limitations.

Reflective-wave N	Displacement error %	Velocity error %	Energy-state error %	All-state target
64	3.641	7.715	6.224	fail
256	3.569	7.819	6.212	fail
1024	3.568	7.813	6.214	fail

Panel	Cluster job	GPU	Scientific source	ROM outliers	Tight FOM outliers
Poisson 2D	3534456	NVIDIA A100 80GB PCIe	085e318ea894	2/1/1	0/0/0
Heat 2D	3529772	NVIDIA A100-PCIE-40GB	7e6d2e39aafd	0/0/0	0/0/0
Burgers 2D	3534502	NVIDIA A100-PCIE-40GB	46ced2fd6bdd	0/0/0	0/0/0
Reflective waves	3534457	NVIDIA A100 80GB PCIe	e1d377928913	0/0/0	0/0/0
Poisson tuning	3548866	NVIDIA A100 80GB PCIe	9528b214d853	see page 2	0/0/0

Outlier counts follow ascending mesh order and include all repetitions. Every panel uses float64 and highest matrix precision, verified GPU execution, burn-in, retained timing arrays, source hashes and checksum-collected full fields. Errors and costs refer to the same solver invocation. Medians pool equal repeats per case; every outlier remains included.

Glossary

Term	Plain-language meaning
NMROM / ROM / FOM	Nonlinear-manifold reduced model / reduced model / full-grid numerical solver.
N / ms / FOM/ROM	Intervals along each axis / milliseconds / ratio of paired median query times.
Current / fixed-initial relative error	Field error divided by the reference field norm at the same time / by the fixed initial norm. The field norm is a spatial L^2 norm.
Worst / adjusted error / target	Maximum over every case, output and repeat / includes empirical reference-refinement allowance / physical error plus original solver and refinement checks.
Stationarity / residual / stall	Small objective gradient / mismatch in the equation / updates or improvements too small to continue, which need not imply stationarity.
CG / BiCGStab / Newton / preconditioner	Iterative linear solver / nonsymmetric linear solver / nonlinear equation iterations / transformation used to help a linear solver.
Crank–Nicolson / midpoint / DST / FFT	Implicit time-stepping formulas / midpoint time stepping / discrete sine transform / fast Fourier transform.
k / r / M / checkpoint / weak modes	Latent dimension / learned spatial-bank size / test-mode count / frozen trained weights / smooth functions used to test the PDE residual.
Development / reference / outlier	Cases already used to select methods / numerical grading solution, with empirical refinement checks / unusually slow retained timing under the native audit’s rule.